

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE II SEMESTER FIRST INTERNAL EXAMINATIONS, APRIL 2024
23-200-204B BASIC ELECTRICAL ENGINEERING (2023 Scheme)
Branch – SFE, ME-A, ME-B, CE-B

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

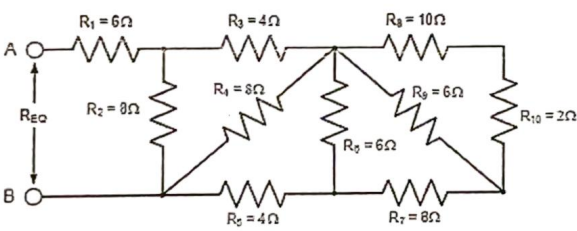
On completion of this course the student will be able to:

CO1	Explain the concepts of various energy sources and electric circuits.
CO2	Apply the basic electrical laws to solve circuits.

PART A

(Answer all questions)

(5 X 2= 10 Marks)

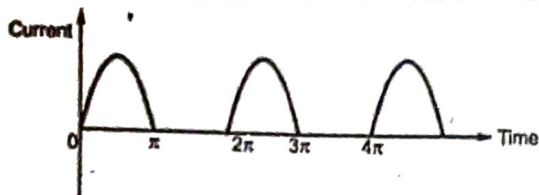
Q.No.	Questions	Marks	BL	CO	PO
I (a)	State and explain Kirchhoff's laws for an electrical circuit.	2	L1	1	1,2
(b)	Obtain the equivalent resistance of the circuit 	2	L3	1	1,2
(c)	The magnetic flux linked with a coil having 250 turns is changed from 1.4 Wb to 2 Wb in 0.45 seconds. Calculate the induced emf in the coil.	2	L3	1	1,2
(d)	State Ohm's law.	2	L1	1	1,2
(e)	Explain Faradays laws of electromagnetic induction.	2	L1	1	1,2

PART B

(Answer the following questions
choosing either the first OR second option)

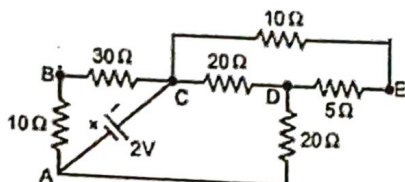
(2 X 10= 20 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Write short notes on induced emfs.	5	L1	1	1,2
(b)	Calculate the RMS value and average value of the given waveform.	5	L2	2	1,2

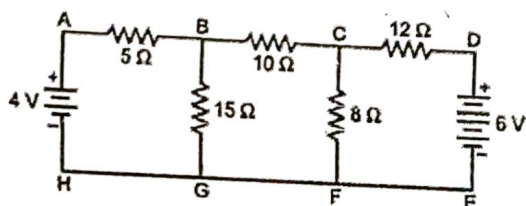


OR

III Using Kirchoff's voltage laws obtain the currents in the circuit shown below. 10 L3 1 1,2



IV Using Kirchoff's current laws obtain the currents in the circuit shown below. 10 L3 1 1,2



OR

V (a) A sinusoidally varying alternating current has an RMS value of 20 A and periodic time of 20 milliseconds. If the waveform of this current enters into its positive half cycle at $t=0$ s, find the instantaneous values of current at $t = 6$ ms and $t = 12$ ms. 5 L4 2 1,2

(b) Define
1. Cycle
2. Frequency
3. Time period
4. Phase difference
5. RMS value 5 L2 2 1,2

Bloom's Taxonomy Levels

L1 - %	L2 - %	L3 - %	L4 - %	L5 - %	L6 - %
22	20	48	10	0	0